

7 Electronic Design

7.1 General Design

The electronic system of Glidesat is divided into 4 boards. 1 mainboard and 4 sub-boards.

1. Mainboard - system control, sensor reading, radio communication, logics
2. Motor Control Board - actuator control
3. Camera Board – high resolution image taking and onboard data storage
4. GPS Module – commercial GPS module

This architecture allows us to separate the control system (mainboard), high speed camera operation, high voltage motor control, and high precision GPS. This separation reduces electrical interference between sub-systems and improves communication reliability. Communication between systems will use following standard interfaces: UART, I2C, SPI, PWM/GPIO.

7.1.1 Design Evolution and Component selections

In the beginning of the project, few MCUs were evaluated. Including the Atmel ATmega series (Arduino equivalent), STM32, ESP32, and RP2040. Atmel chips were rejected since they could not carry out multiple peripherals at the same time reliably, due to the lack of sufficient processing power and memory. STM32 and ESP32 chips were also considered but were found to consume more power, introduce unnecessary complexity, and exceed the performance requirements of our Glidesat application. The RP2040 SoC was selected as it offers dual-core processing power, multiple communication interfaces, low power consumption compared to higher end MCU such as STM32 or ESP32, and great community support and documentation. Although RP2040 is only available as QFN packages, which introduces soldering difficulties, the performance to power ratio advantages justifies the selection.

7.1.2 Board design

Complementary components: Flash memory – W25Q128JVS, Crystal oscillator - X322512MSB4SI, Voltage regulator – AP2112K-3.3

Sensor	Unit
Barometric pressure	BMP280
Temperature	BMP280
IMU (Initial Measurement Unit)	MPU-6050
Battery voltage monitor	GPIO_ADC

To minimise thermal distortion of environmental readings, sensors are placed in separate places with heat generating components, and ventilation opening will be placed near the sensors.

A LoRa transceiver is used for communication. It is connected via SPI and it is used to transmit telemetry data to the ground station. Including: Altitude, temperature, GPS coords, system status, image capture status. For GPS, the Neo-6M GPS module is used.

For camera module, we chose to integrate Raspberry Pi Compute Module 4(CM4) and Arducam 477P high-quality camera module for image capturing. In the beginning of the project, the camera system was planned to be controlled directly by RP2040 via SPI. However, this approach has been rejected since we needed a high resolution(10mp+) camera. The problem was limited RAM of RP2040 (and MCU in general), insufficient bandwidth of SPI for such high-resolution (No high res camera have SPI interface), and lack of native support of the MIPI CSI interface. To capture the high-resolution images, a Linux embedded computer was required. The Raspberry Pi Compute module 4 was therefore selected. The CM4 offers native MIPI CSI interface support, sufficient RAM, stable Linux camera drivers, and data storage via micro-SD card. CM4 will control the camera and trigger it by communicating with the mainboard via UART. There are two possible ways to operate CM4 are evaluated. The first option is operating continuously throughout the mission, and the second option

is turned on when needed using MOSFET. The first option can reduce power cycling and reliability but consume more power. However, the second option consumes less power, but it takes 20-40 seconds of time to boot the Linux system, and it is not highly reliable. Since reliability is a priority in this application, we decided to turn on the CM4 before launching and operate continuously throughout the flight.

The motor board will drive signals for paraglider motors, regulate power for actuators, and have a buzzer as part of the recovery system. The buzzer is added to the motor control board because it requires higher voltage which is already present in the board. Having a separated motor control board allows us to have high current paths separated from the mainboard and prevent electrical noises that might be present with high current paths from affecting sensors or LoRa modules, and it simplifies the PCB Design.

7.1.3 Prototype Mainboard Development (Mainboard Rev.1)

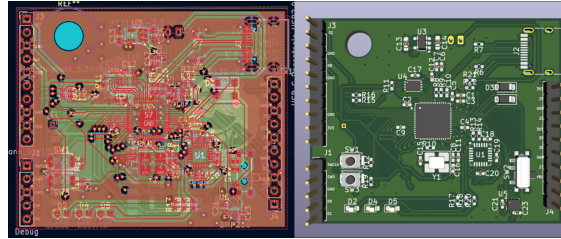


Figure 5: Prototype Mainboard

The first prototype of the mainboard was designed using Kicad, and we are currently waiting for the production. The objectives of Mainboard Rev.1 were to evaluate the general functionality of a designed PCB. Specifically, the RP2040 MCU in multi sensor configuration, verify SPI, I2C, and UART communication interfaces, test power regulation, test radio, etc. We spotted some flaws on PCB after submitting production via JLC PCB, but it should be okay since it is the first prototype.

7.1.4 Prototype Mainboard Development (Mainboard Rev.2)

After finishing Rev.1, we started to work on Rev.2 based on the mistakes we made in Rev.1. We planned pinout of the GPIO pins so we can keep pins organized and planned the board configuration. When we were working on Rev.1, we did not have clear board structure, so we only have a 01x08 JST connector in Rev.1.

Also in Rev.1, we used MPU-6050 as our IMU. But after examining its capabilities, we found that it is not sufficient for controlling parafoil because it lacks a magnetometer which leads to the yaw drift. Thus, we will use 9-DOF IMU for Rev.2. Possibly MPU-9250 and ICM-20948.

7.2 Power Supply

The Glidesat is powered by a rechargeable lithium polymer battery system. Two options were evaluated. Single cell with nominal voltage 3.7V and double cell with 7.4V. Double cell systems offer higher voltage and reduced current for equal power consumption. However, it increases system complexity because of multiple voltage rails, has risk of overvoltage, more weight, and additional safety requirements for charging. Therefore, after evaluation, a single cell LiPo battery is chosen for Glidesat. Single cells are better since most electronics operate at 3.3V (RP2040, sensors, etc.), using a boost converter is simple and efficient, simplified charging, and reduced weight. Estimated power consumption: *CM4* – 2W, *Mainboard* – 1W, *Motor board and buzzer* – 2W so total 5W. For mission of 1h $5W \cdot 1h = 5Wh$ (safety margin of 2 – 5Wh)- therefore, a capacity between 2000mAh and 3000mAh battery is used. DC-DC boost converter generates higher voltage for CM4, actuators, and buzzer. A buck regulator generates 3.3V for RP2040. Battery voltage is monitored by the mainboard ADC.